

Introduction to Quiver

Palantir Foundry's Visual Data Analysis Platform



Purpose and Overview

Quiver is Foundry's point-and-click analysis tool that allows business users and analysts to perform data exploration, transformation, and visualization without writing code. It leverages the Ontology to deliver intelligent, scalable analytics.



Who Uses Quiver

Designed for analysts, operations teams, and decision-makers who require quick insights without needing SQL or coding expertise.



Core Capabilities

Enables self-service analytics through reusable cards and visual canvases that dynamically update with data changes. Supports time series analysis, anomaly detection, and dashboard creation.



Why Quiver Matters

Democratizes analytics across the enterprise—bridging the gap between technical and non-technical users through an intuitive interface.

Quiver Fundamentals

Core Concepts and Architecture



What is Quiver?

A low-code, visual interface for analyzing data modeled in the Palantir Foundry Ontology. Users build analyses by combining reusable 'cards' that represent data processing steps.



When Not to Use Quiver

Avoid for high-volume batch processing, complex ETL pipelines, or use cases requiring custom business logic or extreme performance tuning.



When to Use Quiver

Ideal for ad-hoc analysis, interactive dashboards, anomaly detection, and self-service analytics—especially where quick iteration and dynamic filtering are key.



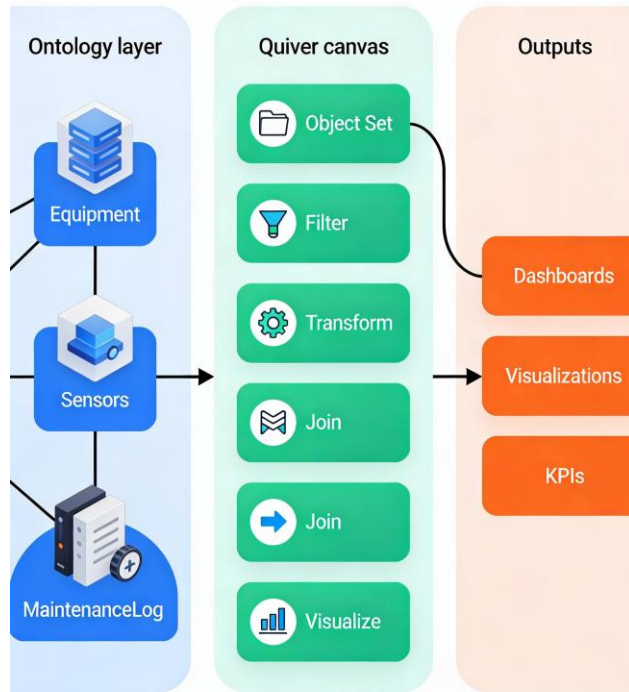
Core Architecture

Built on three pillars—Ontology (semantic model), Cards (data operations), and Canvas (visual workspace)—enabling automatic join path discovery and type-safe data exploration.

Creating and Managing Analyses

Building Data Workflows in Quiver

- **Creating a New Analysis:** Users begin from Foundry's sidebar, select Quiver under Analytics & Operations, and create a new analysis. The first step is adding data by selecting an object type from the Ontology.
- **Object Set Cards:** Each analysis starts with an object set representing domain entities. Object set cards display previews, counts, and allow users to pop out objects, time series, or metrics for exploration.
- **Canvases for Organization:** Analyses can include multiple canvases, which act as logical tabs grouping related workflows. Inactive canvases and hidden plots reduce compute costs.
- **Global Load Settings:** Users can configure whether all cards load fully or only as they come into view, optimizing performance in large analyses.



Core Card Types and Workflow

The Building Blocks of Quiver Analysis



Object-Level Cards

Operate on object sets to filter or navigate relationships. Enable property-based or linked-object filtering with logical operators and thresholds.



Materialization Cards

Execute server-side computations for high-performance analytics. Ideal for large datasets or when joins and aggregations are needed at scale.



Transformation Cards

Perform spreadsheet-like row-wise operations, including numeric formulas and text manipulation. Best suited for datasets under 50,000 rows.



Aggregation and Join Cards

Support metric computation and data merging using count, sum, average, and join operations across related object sets.

Time Series Analysis

Managing and Interpreting Temporal Data in Quiver



Adding Time Series Data

Users can add time series by selecting objects with temporal properties (e.g., Temperature). Quiver automatically renders interactive line charts.



Transformations and Operations

Supports rolling aggregates, derivatives, resampling, cumulative sums, and time range filters for trend analysis and noise smoothing.



Performance Optimization

Quiver uses 1,000 default buckets per series; zooming or pre-filtering increases granularity. Lazy hydration ensures efficient querying.



Anomaly Detection

Time series search cards detect thresholds, deviations, or custom formula-based anomalies—automatically generating event sets for review.

Visualization and Analytics

Turning Data into Actionable Insights



Chart Types

Quiver supports bar, pie, scatter, line, table, and event plots. Each visualization has unique aggregation semantics suited for different analytical contexts.



Advanced Aggregations

Supports formula-based metrics combining multiple aggregations, enabling complex KPI calculations directly within the visualization layer.



Interactivity and Drill-Down

Charts can filter downstream analyses dynamically—clicking a chart element applies context-aware filters across dependent cards.



Dependency Graph

Visualizes how data flows between cards, maintaining analysis integrity even when intermediate cards are removed.

Visual Functions

Reusable, Parameterized Analytics in Quiver



Purpose of Visual Functions

Encapsulate analysis logic into reusable, configurable components that can be applied across analyses without writing code.



Creation Workflow

Build analysis → Select input/output cards →
Configure parameters → Publish as visual function
with metadata and versioning.



Structure and Constraints

Functions include input object sets, processing logic, and a single output. Cannot include visualization cards or nested functions.



Usage and Versioning

Functions can be added to analyses like standard cards, configured with new inputs, and locked to specific versions or set to auto-update.

Parameters for Dynamic Control

Interactive Filtering and Customization in Quiver

- **Purpose of Parameters:** Parameters enable analysts and dashboard users to dynamically modify analyses—changing filters, thresholds, or selected objects without rebuilding workflows.
- **Parameter Types:** Supports Property Value Select, Numeric, Text, and Base parameters, each enabling distinct interactive behaviors.
- **Building Parameterized Analyses:** Example: filtering flights by selected airline or equipment by plant. Parameters propagate downstream to update all dependent visualizations automatically.
- **Integration with Dashboards:** Parameters appear as dropdowns or input fields in dashboards, providing intuitive controls for end users and supporting dynamic drill-down.

Dashboards and Embedding

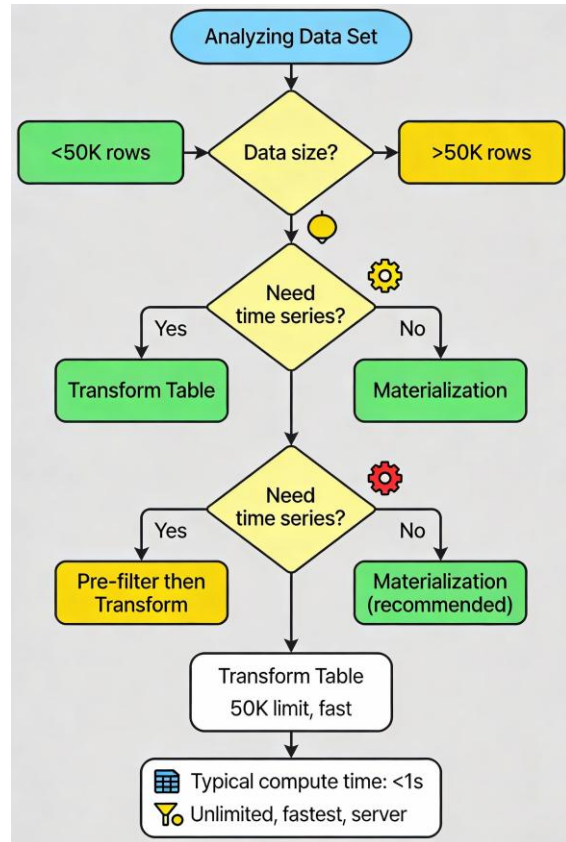
Delivering Insights Through Interactive Views

- **Creating Dashboards:** Analysts curate analyses into dashboards by selecting key cards, organizing tabs, and publishing interactive layouts for end users.
- **Design Principles:** Use tabs over stacked content, position parameters at the top, maintain logical flow, and hide intermediate computation steps for clarity.
- **Inputs and Outputs:** Dashboards can receive inputs (e.g., selected equipment) and expose outputs (e.g., KPIs) to other Foundry apps for integrated workflows.
- **Embedding Options:** Dashboards integrate seamlessly with Workshop, Notepad, and Carbon workspaces, supporting parameter mapping and shared computation caches.

Best Practices and Optimization

Maximizing Efficiency and Scalability in Quiver

- **Canvas and Rendering Strategy:** Split analyses into multiple canvases to improve performance. Use 'Load only visible items' and hide unnecessary plots to reduce computation.
- **Transform Tables vs. Materializations:** Choose Transform Tables for small, time series-enabled datasets (<50K rows) and Materializations for large-scale, high-performance workloads.
- **Time Series Optimization:** Pre-filter large time series, limit non-bucketed operations, and monitor hydration and memory usage for responsiveness.
- **Linked Object Traversal:** Respect object and result limits (100K/10M). For large traversals, pre-compute joins using PySpark and model relationships in Ontology.



Real-World Example: Predictive Maintenance Dashboard

Self-Service Equipment Risk Monitoring in Quiver



Objective

Enable maintenance teams to identify at-risk equipment through visual analytics combining sensor data, anomalies, and maintenance logs.



Data Model

Integrates Equipment, Sensors, and MaintenanceLog objects with time series for temperature, vibration, and pressure to compute KPIs.



Analysis Structure

Four canvases: Overview, Time Series Anomalies, Predictive Risk Score, and Dashboard. Each canvas represents a logical analysis layer.



Key Metrics and Visuals

KPIs: total active equipment, maintenance overdue, average risk score. Visuals include risk distributions, anomaly timelines, and maintenance plans.

Summary and Learning Resources

Empowering Enterprise Analytics with Quiver



Key Takeaways

Quiver democratizes analytics through an intuitive, visual interface powered by Foundry's Ontology, enabling self-service insights at scale.



Workflow Excellence

Effective use of parameters, canvases, and dashboards enhances usability and scalability for complex enterprise analyses.



Core Principles

Success in Quiver relies on ontology-first design, efficient card composition, and disciplined performance optimization.



Learning Resources

Explore Palantir documentation, [Learn.Palantir.com](https://learn.palantir.com) courses, YouTube tutorials, and community engagements for deeper expertise.